

Import Data

```
In [1]: import pandas as pd
```

```
In [2]: data = pd.read_csv(r'D:\Downloads\Udemy-PY\Netflix+Dataset.csv')
```

```
In [3]: data.head()
```

```
Out[3]:
```

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Durat
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0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	August 14, 2020	TV-MA	Seas
1	s2	Movie	07:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, ...	Mexico	December 23, 2016	TV-MA	93 i
2	s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence ...	Singapore	December 20, 2018	R	78 i
3	s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer Connelly...	United States	November 16, 2017	PG-13	80 i
4	s5	Movie	21	Robert Luketic	Jim Sturgess, Kevin Spacey, Kate Bosworth, Aar...	United States	January 1, 2020	PG-13	123 i

1. Find and remove all the duplicate values of dataset

```
In [4]: data.shape
```

Out[4]: (7789, 11)

In [5]: `data[data.duplicated()]`

Out[5]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Du
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6300	s684	Movie	Backfire	Dave Patten	Black Deniro, Byron "Squally" Vinson, Dominic ...	United States	April 5, 2019	TV-MA	
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6622	s6621	Movie	The Lost Okoroshi	Abba T. Makama	Seun Ajayi, Judith Audu, Tope Tedela, Ifu Enna...	Nigeria	September 4, 2020	TV-MA	
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In [6]: `data.drop_duplicates(inplace =True )`

2. Detect all Null values and show them in a heat map

In [7]: `data.isnull()`

Out[7]:

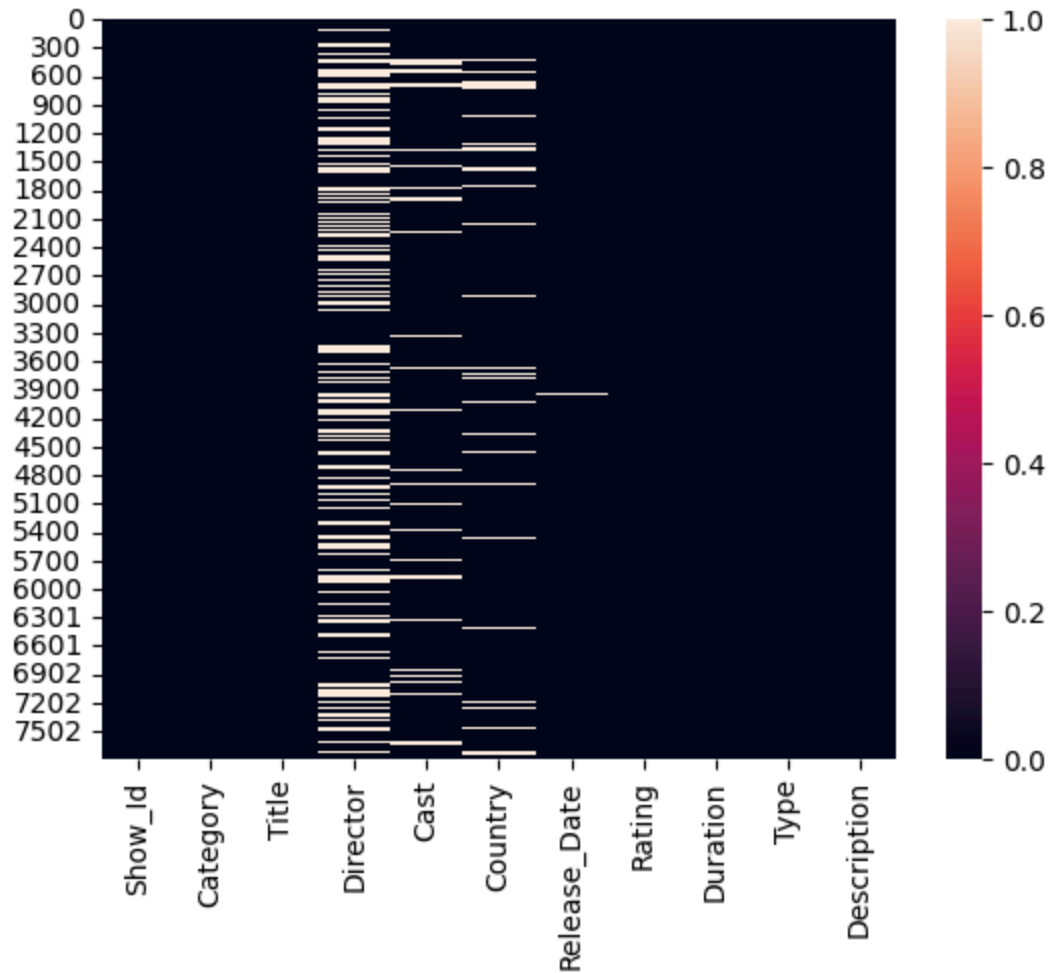
	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Duration
0	False	False	False	True	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...
7784	False	False	False	False	False	False	False	False	False
7785	False	False	False	False	False	False	False	False	False
7786	False	False	False	True	False	True	False	False	False
7787	False	False	False	True	False	False	False	False	False
7788	False	False	False	False	True	False	False	False	False

7787 rows × 11 columns

In [8]: `data.isnull().sum()`

```
Out[8]: Show_Id          0
        Category        0
        Title           0
        Director    2388
        Cast          718
        Country      507
        Release_Date   10
        Rating         7
        Duration       0
        Type           0
        Description    0
        dtype: int64
```

In [9]: `import seaborn as sns`In [10]: `sns.heatmap(data.isnull())`Out[10]: `<Axes: >`



3. For 'House of Cards' what is the Sow Id and who is the director of show?

In [11]: `data[data['Title'].isin(['House of Cards'])]` *#Using isin() function*

Out[11]:

Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Durat
2832	s2833	TV Show	House of Cards	Robin Wright, David Fincher, Gerald McRaney, J...	Kevin Spacey, Robin Wright, Kate Mara, Corey S...	United States	November 2, 2018	TV-MA Seas

Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Durat
2832	s2833	TV Show	House of Cards	Robin Wright, David Fincher, Gerald McRaney, J...	Kevin Spacey, Robin Wright, Kate Mara, Corey S...	United States	November 2, 2018	TV-MA Seas

In [12]: `data[data.Title.str.contains('House of Cards')]` *#Using str.contains function*

Out[12]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Durat
2832	s2833	TV Show	House of Cards	Robin Wright, David Fincher, Gerald McRaney, J...	Kevin Spacey, Robin Wright, Kate Mara, Corey S...	United States	November 2, 2018	TV-MA	Seas



4. In which year highest number of Movies and TV shows were released? Show in graph

In [13]: `data.dtypes`

```
Out[13]: Show_Id      object
Category    object
Title       object
Director    object
Cast        object
Country     object
Release_Date object
Rating      object
Duration    object
Type        object
Description object
dtype: object
```

In [14]: `data['Date_N'] = pd.to_datetime(data['Release_Date'].str.strip(), format='mixed')`In [15]: `data.head()`

Out[15]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Durat
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	August 14, 2020	TV-MA	Seas
1	s2	Movie	07:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, ...	Mexico	December 23, 2016	TV-MA	93 i
2	s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence ...	Singapore	December 20, 2018	R	78 i
3	s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer Connelly...	United States	November 16, 2017	PG-13	80 i
4	s5	Movie	21	Robert Luketic	Jim Sturgess, Kevin Spacey, Kate Bosworth, Aar...	United States	January 1, 2020	PG-13	123 i



In [16]: data['Date_N'].dt.year.value_counts()

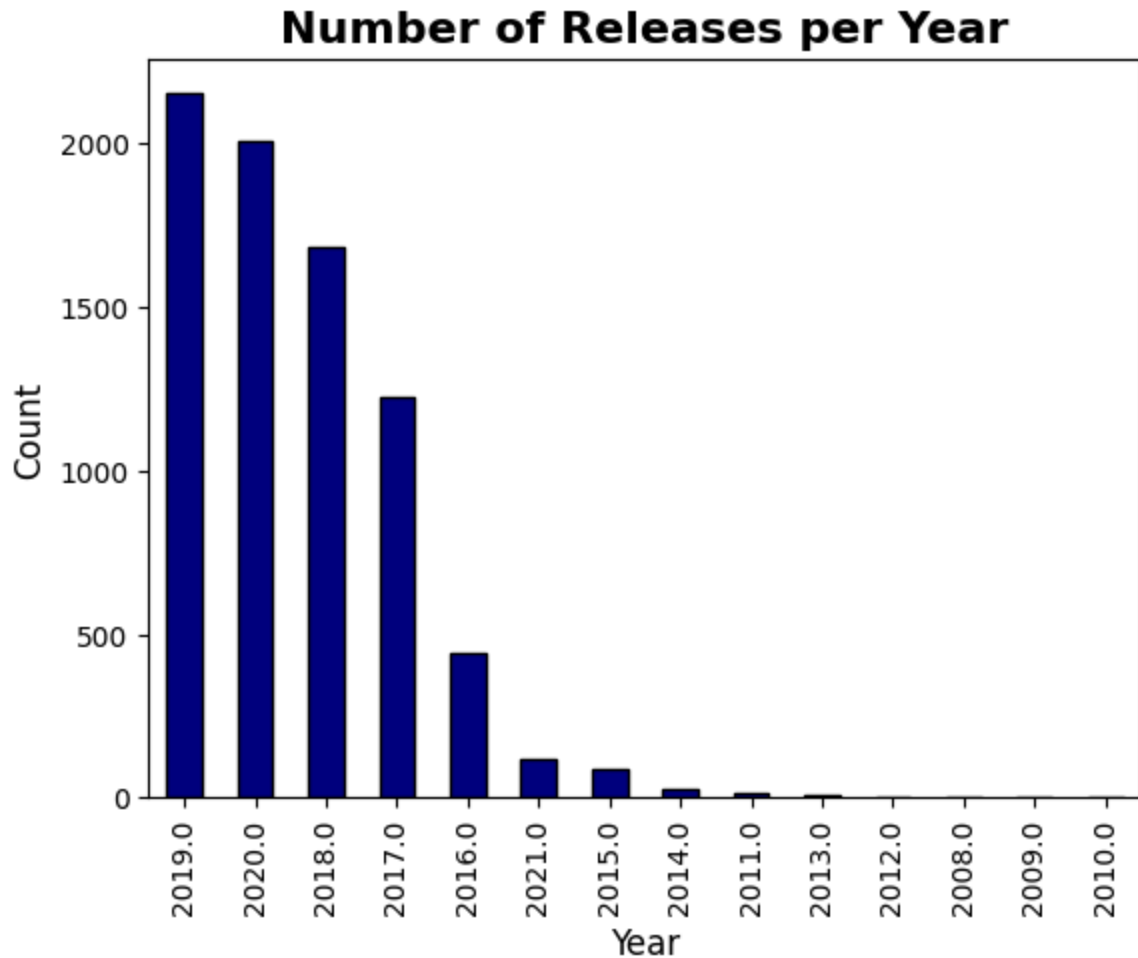
```
Out[16]: Date_N
2019.0    2153
2020.0    2009
2018.0    1685
2017.0    1225
2016.0     443
2021.0     117
2015.0      88
2014.0      25
2011.0      13
2013.0      11
2012.0       3
2008.0       2
2009.0       2
2010.0       1
Name: count, dtype: int64
```

Bar Chart

```
In [17]: # data['Date_N'].dt.year.value_counts().plot(kind = 'bar')
# ---
ax = data['Date_N'].dt.year.value_counts().plot(kind='bar', color='navy', edgecolor=

ax.set_title("Number of Releases per Year", fontsize=16, fontweight='bold')
ax.set_xlabel("Year", fontsize=12)
ax.set_ylabel("Count", fontsize=12)
# ---
# ax = data['Date_N'].dt.year.value_counts().plot(kind='bar', color='navy', edgecol
```

```
Out[17]: Text(0, 0.5, 'Count')
```



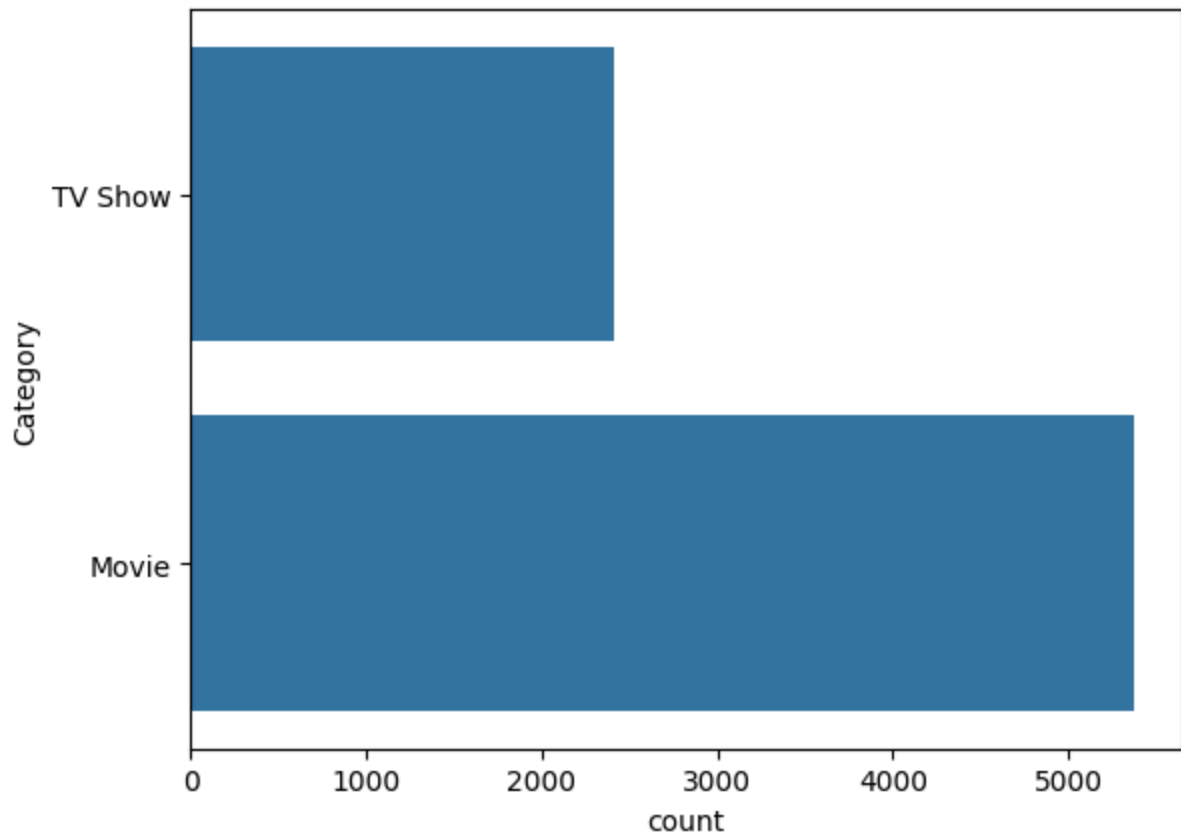
5. How many TV shows and Movies are there in the dataset? Show with the help of bar chart

```
In [18]: data.groupby('Category').Category.count()
```

```
Out[18]: Category
Movie      5377
TV Show    2410
Name: Category, dtype: int64
```

```
In [19]: sns.countplot(data['Category'])
```

```
Out[19]: <Axes: xlabel='count', ylabel='Category'>
```

6. Show all the movies released in the year 2000

```
In [20]: data['Year'] = data['Date_N'].dt.year
```

```
In [21]: data.head(2)
```

Out[21]:	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Duratic
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	August 14, 2020	TV-MA	Season
1	s2	Movie	07:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, ...	Mexico	December 23, 2016	TV-MA	93 m

```
In [22]: data[(data['Category'] == 'Movie') & (data['Year'] == 2020)]
```

Out[22]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	
	4	s5	Movie	21	Robert Luketic	Jim Sturgess, Kevin Spacey, Kate Bosworth, Aar...	United States	January 1, 2020	PG-13
	6	s7	Movie	122	Yasir Al Yasiri	Amina Khalil, Ahmed Dawood, Tarek Lotfy, Ahmed...	Egypt	June 1, 2020	TV-MA
	14	s15	Movie	3022	John Suits	Omar Epps, Kate Walsh, Miranda Cosgrove, Angus...	United States	March 19, 2020	F
	27	s28	Movie	#Alive	Cho Il	Yoo Ah-in, Park Shin-hye	South Korea	September 8, 2020	TV-MA
	28	s29	Movie	#AnneFrank - Parallel Stories	Sabina Fedeli, Anna Migotto	Helen Mirren, Gengher Gatti	Italy	July 1, 2020	TV-14
	...	...	...	...	...	...	...	...	
	7762	s7761	Movie	Zaki Chan	Wael Ihsan	Ahmed Helmy, Yasmin Abdulaziz, Hassan Hosny, H...	Egypt	May 19, 2020	TV-PG
	7783	s7782	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	PG

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating
7784	s7783	Movie	Zozo	Josef Fares	Imad Creidi, Antoinette Turk, Elias Gergi, Car...	Sweden, Czech Republic, United Kingdom, Denmar...	October 19, 2020	TV-MA
7786	s7785	Movie	Zulu Man in Japan	NaN	Nasty C	NaN	September 25, 2020	TV-MA
7788	s7787	Movie	ZZ TOP: THAT LITTLE OL' BAND FROM TEXAS	Sam Dunn	NaN	United Kingdom, Canada, United States	March 1, 2020	TV-MA

1312 rows × 13 columns

7. Show all TV shows and Movies which were released only in India

```
In [23]: data[data['Country'] == 'India']
```

Out[23]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	R
8	s9	Movie	706	Shravan Kumar	Divya Dutta, Atul Kulkarni, Mohan Agashe, Anup...	India	April 1, 2019	
9	s10	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	December 15, 2017	
18	s19	Movie	15-Aug	Swapnaneel Jayakar	Rahul Pethe, Mrunmayee Deshpande, Adinath Koth...	India	March 29, 2019	
20	s21	Movie	Kuch Bheege Alfaaz	Onir	Geetanjali Thapa, Zain Khan Durrani, Shray Rai...	India	September 1, 2018	
21	s22	Movie	Goli Soda 2	Vijay Milton	Samuthirakani, Bharath Seeni, Vinoth, Esakki B...	India	September 15, 2018	
...	...	...	...	...	...	...	...	...
7766	s7765	Movie	Zero	Aanand Rai	Shah Rukh Khan, Anushka Sharma, Katrina Kaif, ...	India	May 21, 2019	
7771	s7770	Movie	Zindagi Na Milegi Dobara	Zoya Akhtar	Hrithik Roshan, Abhay Deol, Farhan Akhtar, Kat...	India	December 15, 2019	
7778	s7777	Movie	Zokkomon	Satyajit Bhatkal	Darsheel Safary, Anupam Kher, Manjari Fadnis, ...	India	November 1, 2018	
7782	s7781	Movie	Zoo	Shlok Sharma	Shashank Arora, Shweta Tripathi, Rahul Kumar, ...	India	July 1, 2018	
7785	s7784	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane	India	March 2, 2019	

Show_Id	Category	Title	Director	Cast	Country	Release_Date	R
				Dias, Raaghav Chanan...			

923 rows × 13 columns

8. List 10 Directors who gave highest no. of Movies and TV shows to the Netflix

```
In [25]: data['Director'].value_counts().head(10)
```

```
Out[25]: Director
Raúl Campos, Jan Suter    18
Marcus Raboy              16
Jay Karas                 14
Cathy Garcia-Molina      13
Youssef Chahine          12
Jay Chapman               12
Martin Scorsese           12
Steven Spielberg         10
David Dhawan              9
Lance Bangs               8
Name: count, dtype: int64
```

9. List all records where Category is 'Movie' and type is 'Comedy' or Country is 'United Kingdom'

```
In [29]: data[(data['Category'] == 'Movie') & (data['Type'] == 'Comedies') | (data['Country']
```

Out[29]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating
33	s34	Movie	#realityhigh	Fernando Lebrija	Nesta Cooper, Kate Walsh, John Michael Higgins...	United States	September 8, 2017	TV-14
72	s73	Movie	17 Again	Burr Steers	Zac Efron, Leslie Mann, Matthew Perry, Thomas ...	United States	January 1, 2011	PG-13
84	s85	Movie	21 & Over	Jon Lucas, Scott Moore	Miles Teller, Skylar Astin, Justin Chon, Sarah...	United States	April 16, 2019	I
148	s149	Movie	A Bad Moms Christmas	Jon Lucas, Scott Moore	Mila Kunis, Kristen Bell, Kathryn Hahn, Susan ...	United States, China	February 14, 2020	I
174	s175	Movie	A Futile and Stupid Gesture	David Wain	Will Forte, Domhnall Gleeson, Martin Mull, Joe...	United States	January 26, 2018	TV MA
...	...	...	...	...	...	...	...	...
7341	s7340	Movie	Undercover Brother 2	Leslie Small	Michael Jai White, Vince Swann, Barry Bostwick...	United States	November 5, 2019	I
7590	s7589	Movie	Where's the Money	Scott Zabielski	Andrew Bachelor, Kat Graham, Mike Epps, Terry ...	United States	January 22, 2018	I

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating
7623	s7622	Movie	Wild Oats	Andy Tennant	Shirley MacLaine, Jessica Lange, Billy Connoll...	Germany, United States	May 13, 2017	PG-13
7634	s7633	Movie	Wine Country	Amy Poehler	Amy Poehler, Maya Rudolph, Ana Gasteyer, Rache...	United States	May 10, 2019	I
7664	s7663	Movie	Work It	Laura Terruso	Sabrina Carpenter, Jordan Fisher, Liza Koshy, ...	United States	August 7, 2020	TV-14

97 rows × 13 columns

10. Show all records in which Tom Cruise was present in Cast

```
In [44]: data_new = data.dropna()
```

```
In [47]: data_new[data_new['Cast'].str.contains('Tom Cruise')]
```

Out[47]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating
3860	s3861	Movie	Magnolia	Paul Thomas Anderson	Jeremy Blackman, Tom Cruise, Melinda Dillon, A...	United States	January 1, 2020	R
5071	s5071	Movie	Rain Man	Barry Levinson	Dustin Hoffman, Tom Cruise, Valeria Golino, Ge...	United States	July 1, 2019	R

11. What are the different Ratings defined by Netflix?

```
In [49]: data['Rating'].unique()
```

```
Out[49]: array(['TV-MA', 'R', 'PG-13', 'TV-14', 'TV-PG', 'NR', 'TV-G', 'TV-Y', nan,
               'TV-Y7', 'PG', 'G', 'NC-17', 'TV-Y7-FV', 'UR'], dtype=object)
```

```
In [50]: data['Rating'].nunique()
```

```
Out[50]: 14
```

11.a How many movies got TV-4 Rating in Canada?

```
In [52]: data[(data['Rating'] == 'TV-4') & (data['Country'] == 'Canada')].shape
```

```
Out[52]: (0, 13)
```

11.b How many TV Shows got 'R' Rating after 2018?

```
In [58]: len(data[(data['Rating'] == 'R') & (data['Year'] > 2018)])
```

```
Out[58]: 450
```

12. What is the Max Duration of Netflix Movie/Show?

```
In [64]: data[['Minute', 'Unit']] = data['Duration'].str.split(' ', expand = True)
```

```
In [66]: data.head(1)
```

```
Out[66]:
```

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Duration
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	August 14, 2020	TV-MA	Season



```
In [72]: data.loc[data['Minute'].idxmax()]
```



```
Out[72]: Show_Id          s28
         Category        Movie
         Title           #Alive
         Director        Cho Il
         Cast            Yoo Ah-in, Park Shin-hye
         Country          South Korea
         Release_Date     September 8, 2020
         Rating           TV-MA
         Duration         99 min
         Type             Horror Movies, International Movies, Thrillers
         Description      As a grisly virus rampages a city, a lone man ...
         Date_N           2020-09-08 00:00:00
         Year             2020.0
         Minute           99
         Unit             min
         Name: 27, dtype: object
```

13. Which individual Country has Max no. of Movies

```
In [74]: data_Movie = data[data['Category'] == 'Movie']
```

```
In [78]: data_Movie.head()
```

Out[78]:

	Show_Id	Category	Title	Director	Cast	Country	Release_Date	Rating	Duratic
1	s2	Movie	07:19	Jorge Michel Grau	Demián Bichir, Héctor Bonilla, Oscar Serrano, ...	Mexico	December 23, 2016	TV-MA	93 m
2	s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hii, Lawrence ...	Singapore	December 20, 2018	R	78 m
3	s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer Connelly...	United States	November 16, 2017	PG-13	80 m
4	s5	Movie	21	Robert Luketic	Jim Sturgess, Kevin Spacey, Kate Bosworth, Aar...	United States	January 1, 2020	PG-13	123 m
6	s7	Movie	122	Yasir Al Yasiri	Amina Khalil, Ahmed Dawood, Tarek Lotfy, Ahmed...	Egypt	June 1, 2020	TV-MA	95 m

In [80]: `data_Movie['Country'].value_counts()`

```
Out[80]: Country
United States      1850
India              852
United Kingdom     193
Canada            118
Egypt              89
...
Egypt, Austria, United States    1
Russia, Spain                   1
Croatia, Slovenia, Serbia, Montenegro  1
United Arab Emirates, Jordan      1
Belgium, United Kingdom, United States  1
Name: count, Length: 590, dtype: int64
```

14. Sort the Dataset by Year of Releases

```
In [77]: data['Year'].sort_values(ascending=True).head(10)
```

```
Out[77]: 1765    2008.0
7115    2008.0
5766    2009.0
3248    2009.0
3840    2010.0
2041    2011.0
7232    2011.0
6847    2011.0
2622    2011.0
5042    2011.0
Name: Year, dtype: float64
```